w11L51: W11L1: Text Summarization - LexRank

Text Summarization

What is a summary?

A summary is a text that is produced from one or more texts, that contains a **significant portion of the information** in the original text(s), and that is **no longer than half** of the original text(s). (Hovy, 2008)

What is text summarization?

Text summarization is the process of **distilling** the most important information from a source (or sources) to produce an **abridged** version for a particular *user* or *task*. (Mani and MayBury, 2001)

Humans have an incredible capacity to condense information down to the **critical bit.**

- *Calvin Coolidge (heard lecture for 4 hours), on being asked what a clergyman preaching on sin said.*
    - *"He said he is against it."*

Automatic Text Summarization

Goal of a Text Summarization System

To give an **overview** of the original document in a **shorter** period of time.

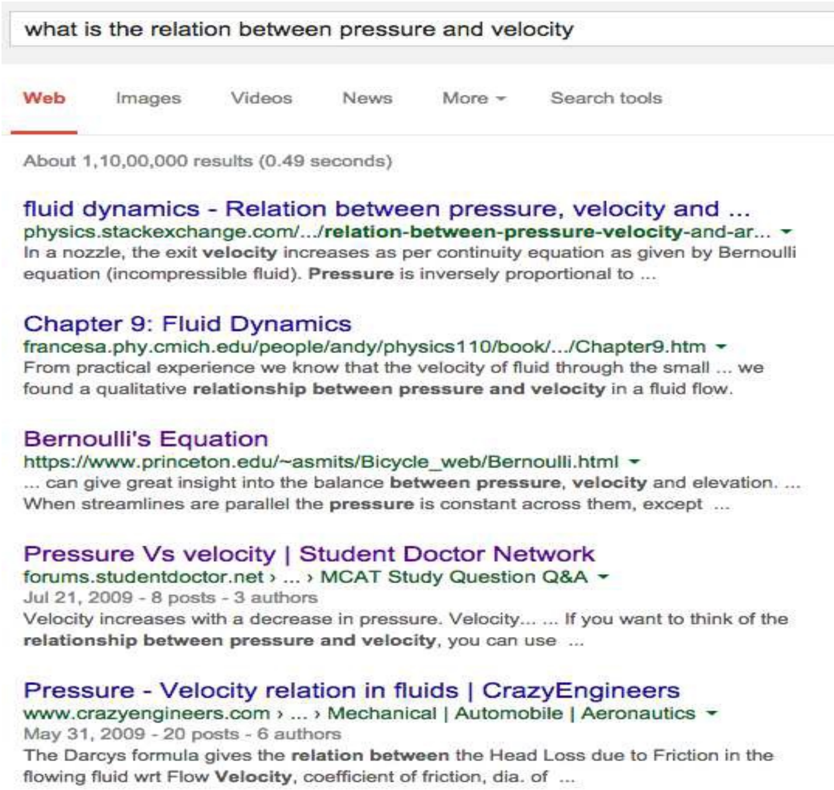
Summarization Applications

- ***outlines*** *or* ***abstracts*** of any document, news article etc.
- ***summaries*** of email threads
- ***action items*** from a meeting
- ***simplifying*** text by compressing sentences

Application: Generating Snippets

Robert O'Neill taking credit for killing Osama bin Laden sparks debate
Hindustan Times - 1 hour ago
Some special operations service members and veterans are unhappy that one of their own has taken credit publicly for killing Osama bin Laden.

It's been special knock as wait has been long: Rayudu
Hindustan Times - 1 hour ago
An elated Ambati Rayudu said Friday that his maiden hundred in international cricket will certainly be a "special one" as it took a long time to come.

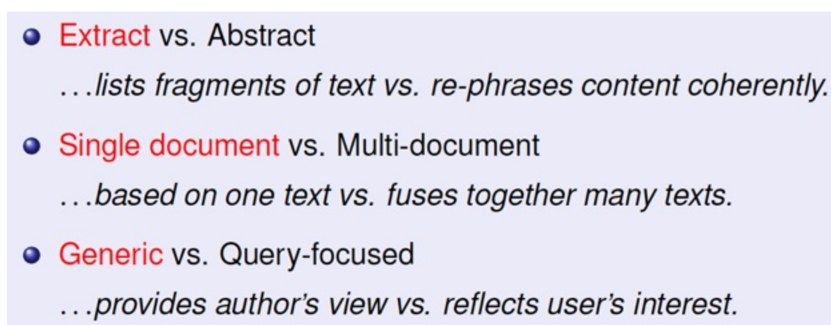


you will get snippets

Automatic Text Summarization

Genres of Summary

Extract vs. **Abstract**
. . .lists fragments of text vs. re-phrases content coherently.
Single document vs. Multi-document
. . .based on one text vs. fuses together many texts.
Generic vs. Query-focused
. . .provides author's view vs. reflects user's interest.
… like complex Q&A task



Will talk about Extractive for single document
Can use for Abstractive and multiple documents

Query-focused summarization can be thought of as a *complex question answering system*

Summarization: Main stages

1. Content Selection

- Choose sentences to extract from the document - find important sentences

2. Information Ordering

- Choose an order to place them in the summary
- how will i make it more readable presentable
- How it can be more coherent

3. Sentence realization

- Simplify the sentences

4. Removing Redundancy

- Increase **diversification** by removing redundant sentences

The most **basic algorithm** only does the first stage, **content selection.**

Unsupervised content selection; Luhn (1958)

Intuition

Choose **sentences** that have **salient** or **informative words** (what is the information content)

1. TF-IDF weight

Two approaches to define salient words

- *tf-idf:* weigh each word w_i in document j by tf-idf

$$weight(w_i) = tf_{ij} \times idf_i$$

2. What are domain specific words - take sentences which have **topic signature**

- *Topic signatures:* choose a smaller set of salient words, specific to that domain

$$weight(w_i) = 1 \text{ if } w_i \text{ is a specific term (use mutual information)}$$

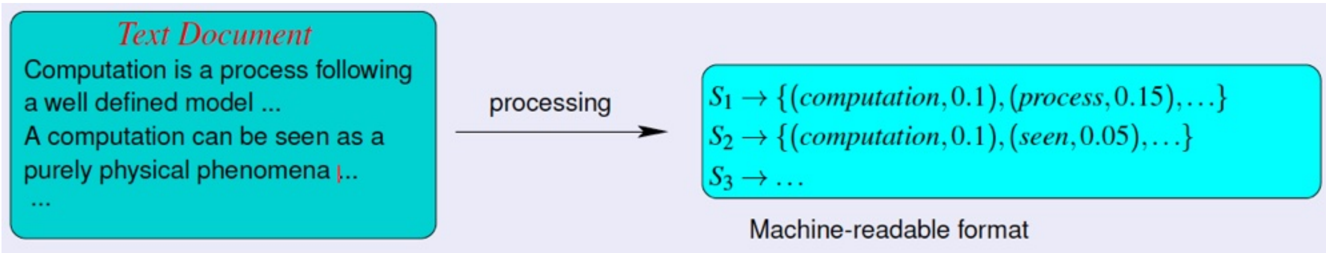
Based on weights from one of above two approaches - take average

Weighing a sentence

$$weight(s) = \frac{1}{|S|} \sum_{w \in S} weight(w)$$

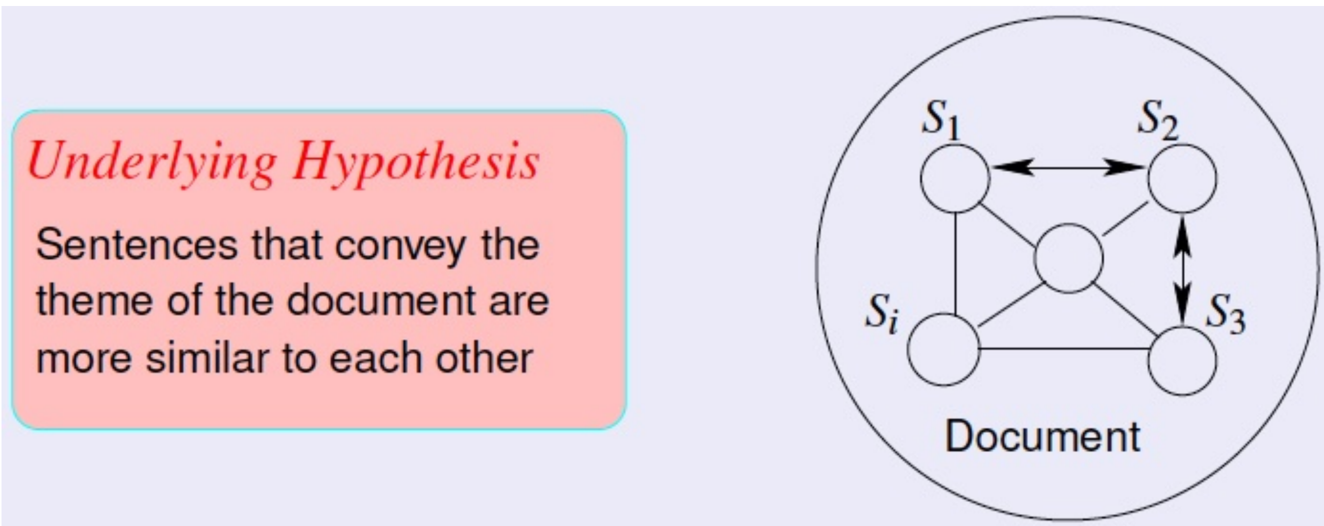
LexRank: A Graph-based approach

- Principled way of approach
-
1. *Document Representation*



Assign tf-idf eg. 0.1, 0.15, 0.05 to each sentence

2. Finding the most salient sentences (Assignment question)



What is the cosine similarity between sentences?

Hypothesis: it is about underlying theme

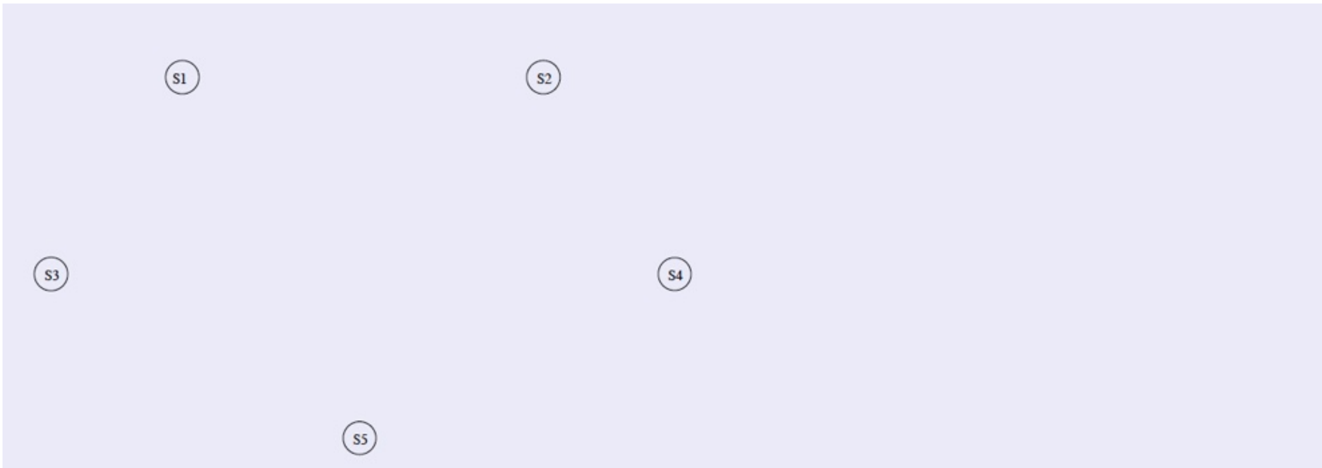
- *Sentences should be talking about **theme***
- *This sentence should be similar to many other sentences*
    - *It will be important*
    - *Other important sentences should also be relevant*

In Page rank - if many pages are linked to it

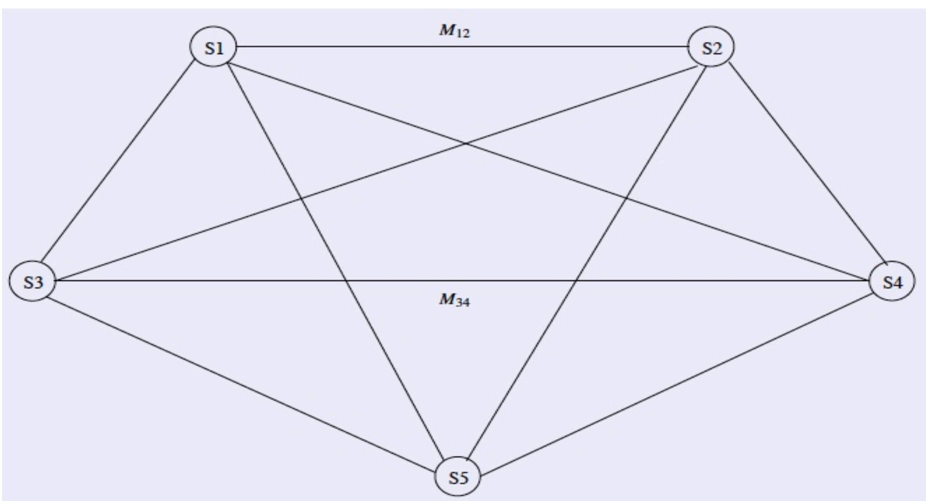
Sentence Centrality Measure

Finding the most salient sentences

1. A document graph is constructed with sentences as the vertices



2. A sentence similarity function is used to calculate the edge weights.



$$\tilde{M} = \begin{bmatrix} 0.0 & 0.5 & 0.0 & 0.4 & 0.1 \\ 0.5 & 0.0 & 0.5 & 0.0 & 0.0 \\ 0.0 & 0.5 & 0.0 & 0.5 & 0.0 \\ 0.4 & 0.0 & 0.4 & 0.0 & 0.2 \\ 0.3 & 0.0 & 0.0 & 0.7 & 0.0 \end{bmatrix}$$

N sentences: N x N pairs

3. PageRank based algorithm is used to compute the sentence centrality vector *I.*

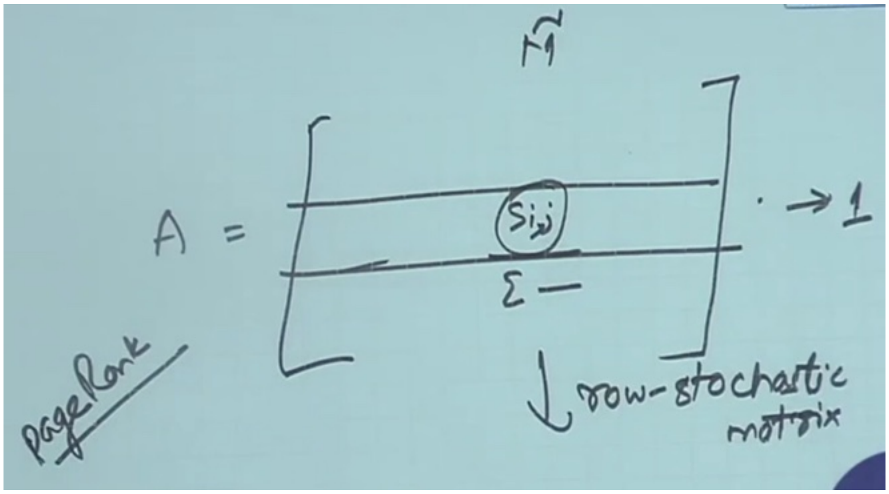
Informative vectors

$$I_j = \mu \cdot \sum_{\forall k \neq j} I_k \cdot \tilde{M}_{k,j} + \frac{1-\mu}{|S|}$$

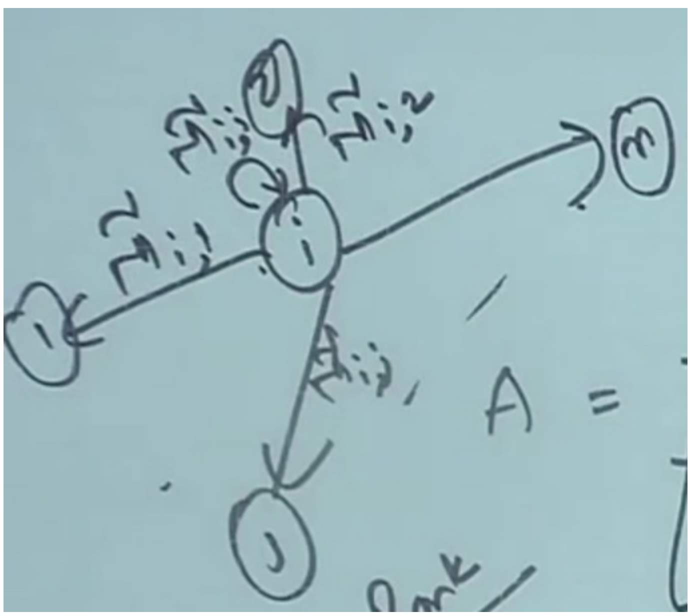
$$I = \begin{bmatrix} 0.22 & 0.18 & 0.2 & 0.3 & 0.1 \end{bmatrix}$$

Node: that has highest weight is - most important

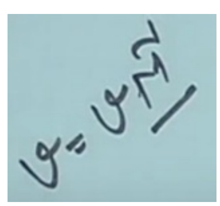
19min: Sij - how similar two sentences are



Convert it to **row-stochastic matrix** - M-tilda
- All elements in individual rows sum to 1
- How does it help?
    - Gives probability - how can we see it as probability
    - Going from node i to node j

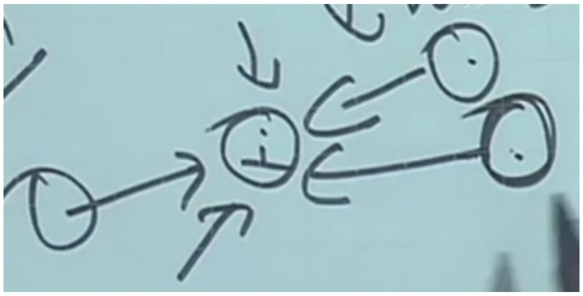


Doing random walk, you can go from node i to other nodes

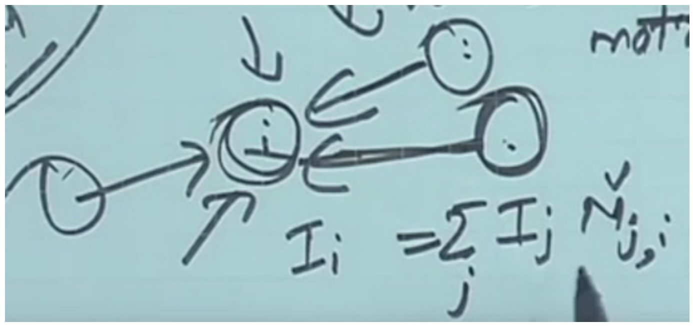
When you do random walk, it converges to 
V is page rank vector

A web page is important if many pages link to it

- How do they correlate?



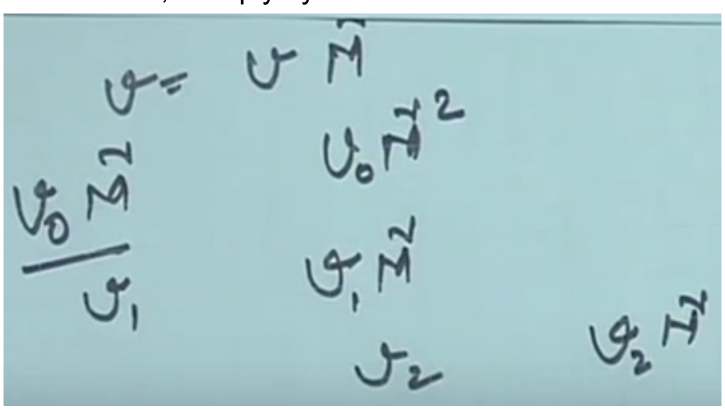
you will stay at important node for longer time

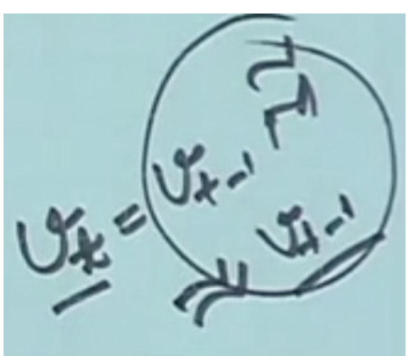
As equation: importance of node 

Importance of node = importance of other nodes * prob of linking from that node

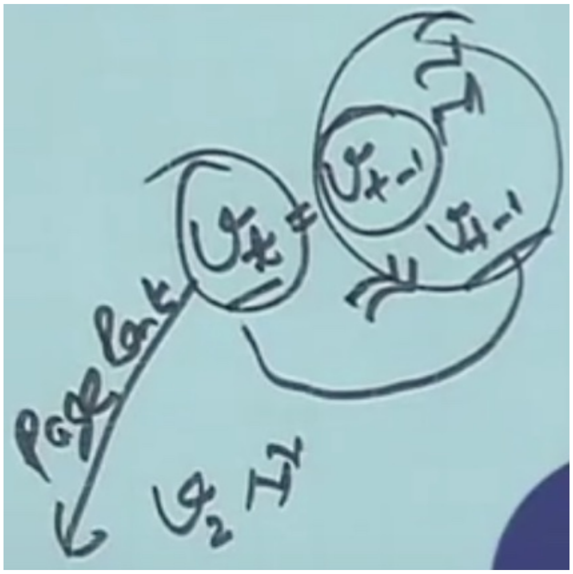
Iteratively compute all I values

Start with v0, multiply by M-tilda/v1 and so on

 it will converge

 where vt and vt-1 are same

Keep on multiplying by M-tilda -> find the difference -> at sometime below threshold

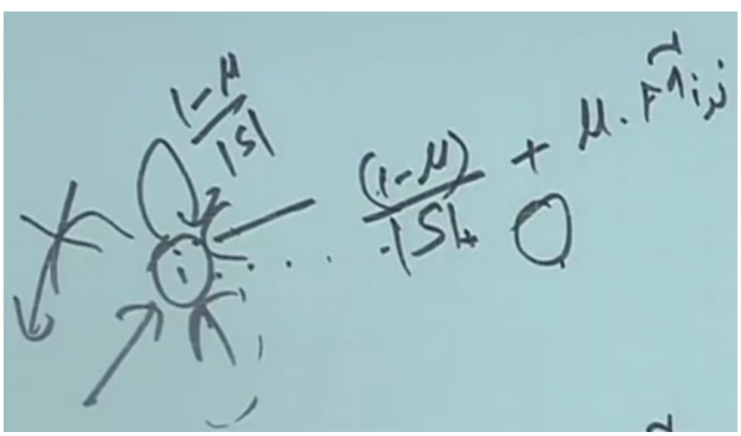


$$I_j = \mu \cdot \sum_{\forall k \neq j} I_k \cdot \tilde{M}_{k,j} + \frac{1-\mu}{|S|}$$

$$I = \begin{bmatrix} 0.22 & 0.18 & 0.2 & 0.3 & 0.1 \end{bmatrix}$$

Actual formula

What is
S - no. of sentences
mu is small teleport probability'



No out going node - random walk is stalled

- Assign small prob to all nodes - which is (1-mu)/S going to all node
- To balance have first term mu * M-tilda-ij

Mu is 0.85 - roughly taken

You will find page rank values I = [0.22   0.18   0.2   0.3   0.1]

- 4th sentence has highest prob (0.3)
- If you want to take longer take 1st (0.22 prob) or 3rd (0.2 prob)
- Take one of them as required

For diversity

- Give less weight to a sentence which is similar to other sentence

Removing Redundant Sentences

Maximal Marginal Relevance

- *How much more relevance are you brining in*

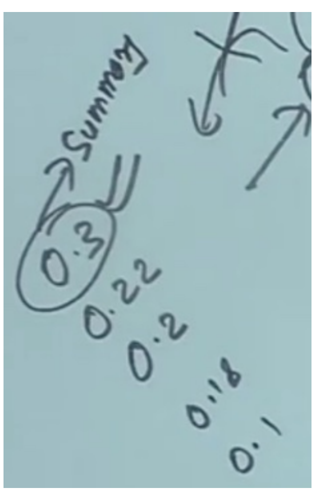
- An iterative method for content selection from a selected list of important sentences

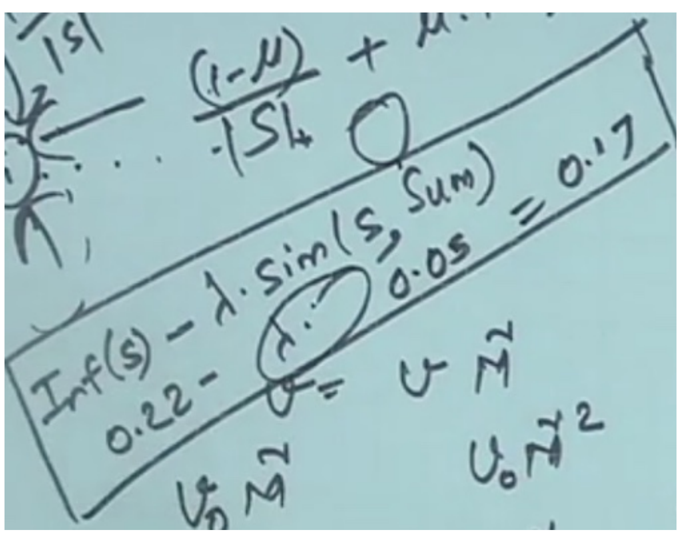
- Iteratively choose the best sentence to insert in the summary that is minimally redundant with the summary so far (Sum)

$$Inf(s)_{MMR} = max_{s \in D}(Inf(s) - \lambda \cdot sim(s, Sum))$$

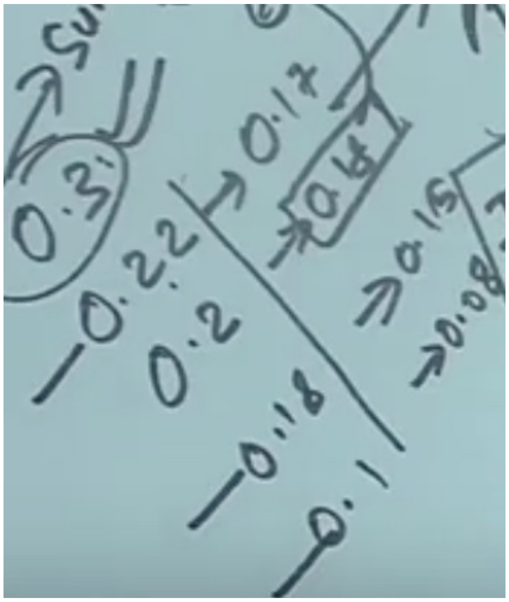
where $Inf(s)$ denotes the informativeness score of a sentence

- Minimally redundant with summary so far





Again rerank -> informative score - lambda similarity
Do for all sentences



Suppose it comes out to be
Sort and take one with highest - 0.4

Now you have two sentences
Again compute - take highest
- Until you get desired no. of sentences
- You have diversity

w11L52: W11L2: Optimization Based Approaches for Summarization

Discussed:
Graph based method - LexRank - take sentence which has high PageRank
Find diverse sentence - MMR - Max Margin Relevance algorithm

- Sort and chose sentence which has highest value
- Doesn't give flexibility in how you want to choose

You can modify it for different tasks: give flexibility to

- Define own optimization function
- Add constraints

Global Inference

$$D = t_1, t_2, \ldots, t_{n-1}, t_n$$

Let us **define document** D with t*n* textual units
Textual units are sentences - n sentences

- Let $Rel(i)$ be the relevance of t_i to be in the summary
- Let $Red(i,j)$ be the redundancy between t_i and t_j
- Let $l(i)$ be the length of t_i

Relevance:

- weights - do average
- Find page rank
- How many named entities are there
- What is the redundancy - we don't want multiple diverse sentences in summary
    - Cosine similarity
    - How much apart in the document - near, far apart
    - No. of words in text

Inference Problem

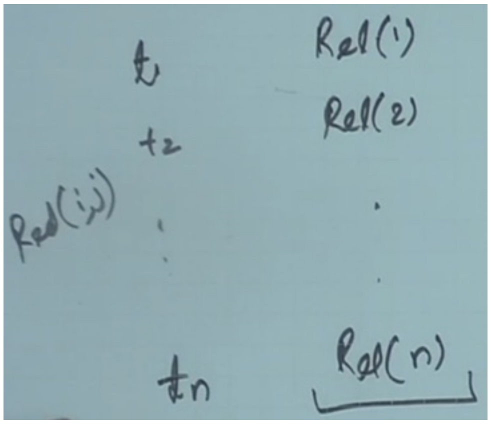
- The inference problem is to select a subset S of textual units from D such that summary score of S, i.e., $s(S)$, is maximized.
- $S = \arg\max_{S \subseteq D} \left[\sum_{t_i \in S} Rel(i) - \sum_{t_i, t_j \in S, i<j} Red(i,j) \right]$
  such that $\sum_{t_i \in S} l(i) \leq K$, where k denotes the maximum length of the summary

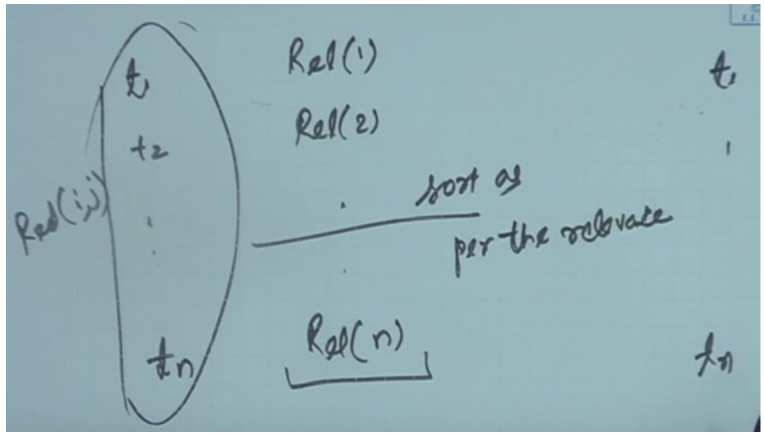
Want summary of length k

- What should be optimisation criteria
- Select subset whose score is maximum
- [relevance - redundancy]
- Will not repeat pair so i < j
- Select subset which maximises it
- **Max rel, Min redundancy (Assignment question 1)**
- Constraint: K ma length of summary
- How do you achieve this - maximise optimisation

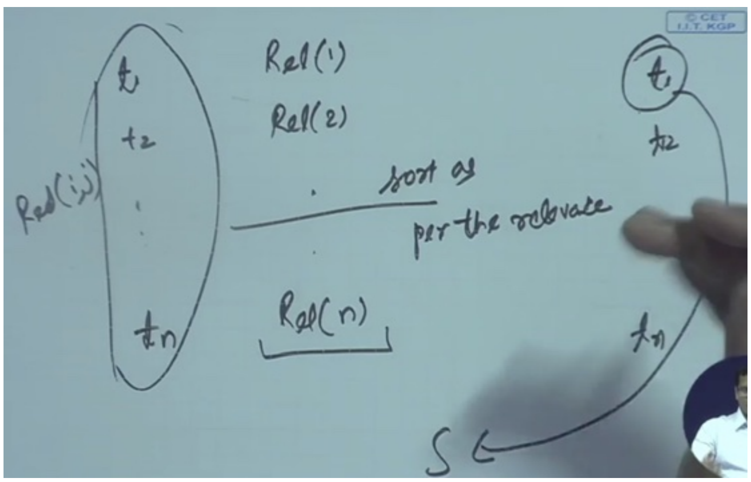
A Greedy Solution

1. Sort D so that $Rel(i) > Rel(i+1) \forall i$
2. $S = \{t_1\}$
3. while $\sum_{t_i \in S} l(i) < K$
4. $\quad t_j = \arg\max_{t_j \in D-S} s(S \cup \{t_j\})$
5. $\quad S = S \cup \{t_j\}$
6. return S



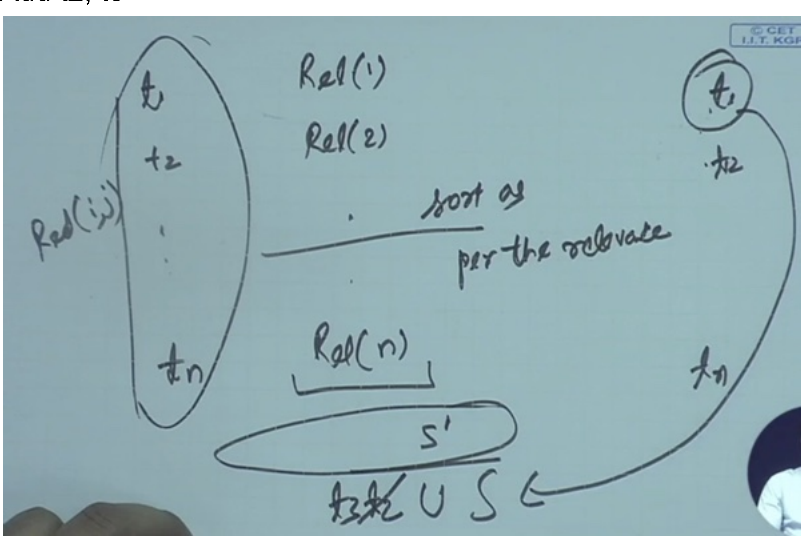


first sort t1 to tn, assume order is still t1 to tn

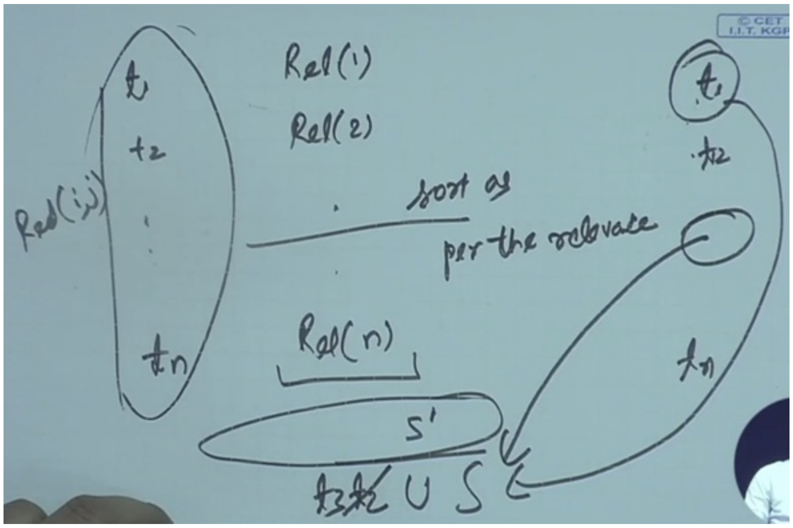


Take t1 in summary. Sum of rel t1 and rel t2 minus - (red of t1 and t2)

Add t2, t3



Again one by one add to Summary



Add one by one which one gives highest score

1. Sort D so that $Rel(i) > Rel(i+1) \forall i$
2. $S = \{t_1\}$
3. while $\sum_{t_i \in S} l(i) < K$
4. $\quad t_j = \arg\max_{t_j \in D-S} s(S \cup \{t_j\})$
5. $\quad S = S \cup \{t_j\}$
6. return S

3 - until desired length
4 - take remaining sentences - add one which has highest score

Achieved both relevance and diversity

- It is greedy approach
- Not going to an optimal solution - near optimal
- Good enough for decent solution

You can use Dynamic Programming

- Storing results, what is the best way of achieving summary 5, 10, 15

Integer Linear Programming (ILP)

- Greedy algorithm (previously) is an approximate solution
- Use **exact** solution algorithms with ILP - optimal value
- ILP is a **constrained optimization problem**
  - E.g. Summation less than K
- Many solvers on the web - you can take any one of them
- To understand
  - Define the **constraints** based on **relevance** and **redundancy** for summarization

Sentence Level ILP Formulation

Optimization Function

$$\text{maximize } \sum_i \alpha_i Rel(i) - \sum_{i<j} \alpha_{ij} Red(i,j)$$

We are not taking subsets

$$\sum_i \underbrace{\alpha_i}_{\{0,1\}} \underline{Rel(i)} - \sum_{i,j} \underbrace{\alpha_{ij}}_{\{0,1\}} \underline{Red(i,j)}$$

alpha-i : whether ith unit is included in summary or not - should have value of 0 or 1

- 0 not in summary, 1 in summary

alpha-ij : can take value 0 or 1

- ij both are in summary 1

$$\alpha_i = 1 \ \forall i \qquad \alpha_{ij} = 0 \ \forall i,j$$

Can put solution, you are taking whole document

Constraints

Length of summary less than K - $\sum \alpha_i l(i) \le K$

Will ensure this solution is not acceptable $\cancel{\alpha_i = 1 \ \forall i}$

Whenever alpha-i to be 1 and alpha-ij =0 - don’t want

- Also don’t want alpha-ij = 1 and alpha-i to be 0

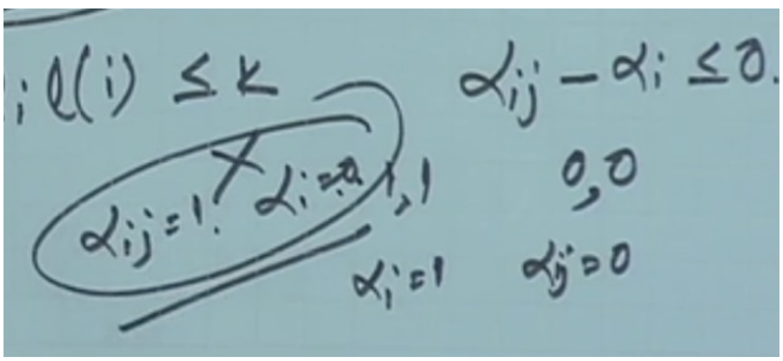
When alpha-i and alpha-j are 1, want alpha-ij has to be 1

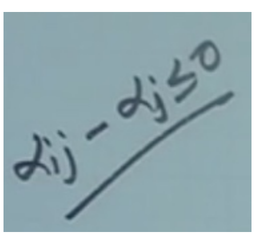
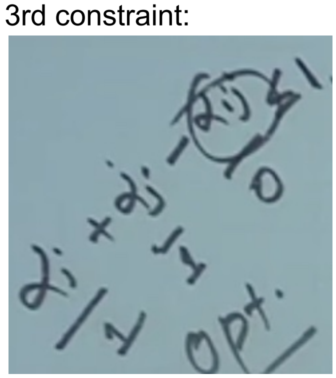
- Need some more constraints

How do we put things in constraint

such that $\forall i, j$:

- $\alpha_i, \alpha_{ij} \in \{0, 1\}$
- $\sum_i \alpha_i l(i) \leq K$
- $\alpha_{ij} - \alpha_i \leq 0$
- $\alpha_{ij} - \alpha_j \leq 0$
- $\alpha_i + \alpha_j - \alpha_{ij} \leq 1$

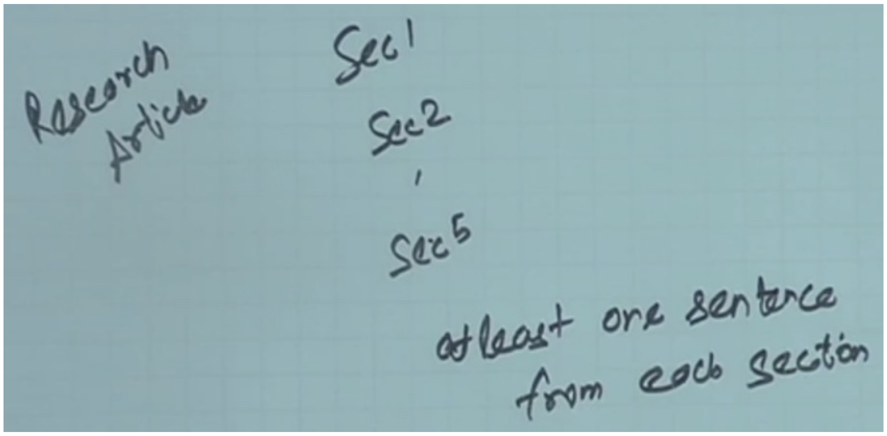


 understand same way	3rd constraint:  1 or 1 or 0 <=1 Can’t have -ij = 0 when a-i and a-j are one

Now putting constraints in terms of Integer Linear Programming

Is generic enough
Depending on your task, you can define your own optimization function and constrains.

Doing summary from a research article:



- Has sections
- In summary want at least one sentence from each section
    - Define section information

News article - Data is temporal

- If two information are in two time point, take information from later time point than previous time point - certain time period more than other time period
- Optimization fn - put weight (importance) to different days
- Max multiple things together
- Still be able to find solution using ILP

2. The next steps: Sentence Ordering

- Heuristic approach

Chronological ordering: the simplest method

1. List the sentences in the order, they appear in the document

- First sentence you will put first

Coherence

2. Choose orderings that make neighboring sentences **similar** (by cosine)
3. Choose orderings in which neighboring sentences discuss the **same entity, events**

Topical ordering

4. Learn the ordering of **topics** in the source documents

3. The next steps: Simplifying Sentences

- Can get more space

Parse sentences, use rules to decide which modifiers to **prune**

- **Initial adverbials:** *For example, on the other hand, as a matter of fact, at this point, …*
  - *Not important to convey information*
- **PPs without named entities:**
  - The commercial fishing restrictions in Washington will not be lifted unless the salmon population increases [*PP to a sustainable number*] *Not important to convey information*
- **Attribution clauses:**
  - Rebels agreed to talks with government officials, *international observers said Tuesday*
- **Appositives:**
  - Rajan, *28, an artist who was living at the time in Philadelphia*, found the inspiration in the back of city magazines

These rules can be given to the system, or learn the rules (by inputs provided by humans)

w11L53: W11L3: Summarization: Evaluation (20min)

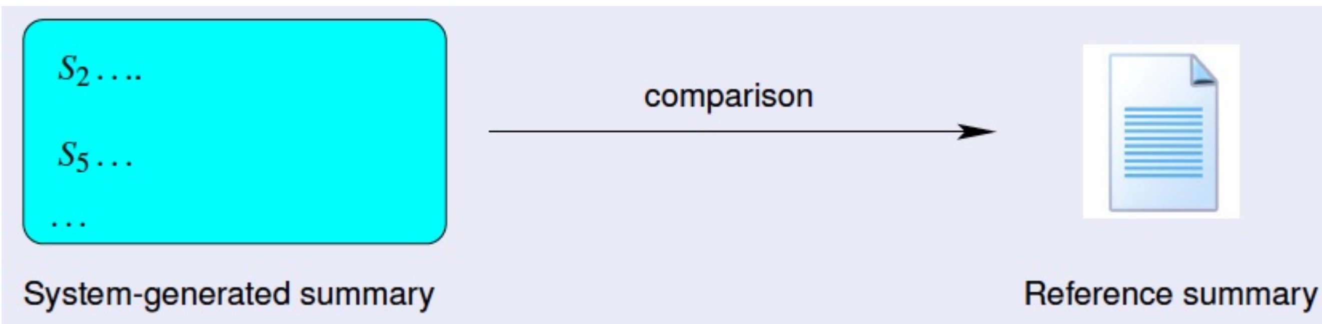
System Evaluation

- Human should say what is the good information coverage (on scale 1-5)
- What kind of diversity (on scale 1-5)
- Readable - coherent (on scale 1-5)

Can you do it automatically?

- Compare system generated summary with Reference summary

Evaluation Criteria ***System Evaluation***



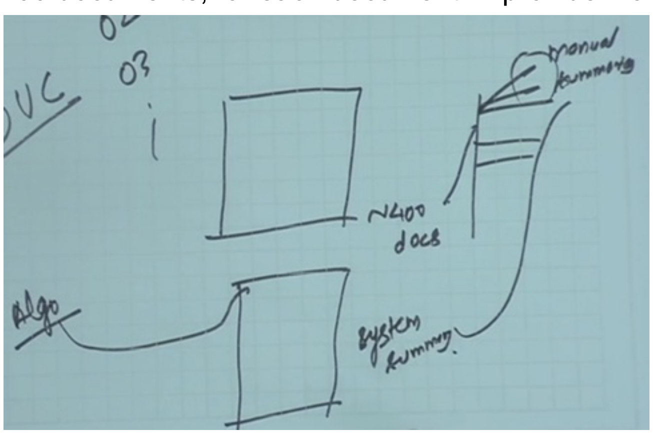
ROUGE

Recall Oriented Understudy for Gisting Evaluation

- *Not as good as human evaluation but much more **convenient***
- *Human evaluation may vary*

Toolkit available for download.

DUC (o2, o3 datasets): Document Understanding Conferences

- 400 documents, for each document -> provide manual summary
- 

vary the algorithm and compare

ROUGE for evaluation (has many variations - unigram, bigram, longest common sub-sentence)
Popular - ROGUE-1 (unigram) ROGUE-2 (birgram)

Given a document D, and an automatic summary X:

- Have N humans produce a set of reference summaries of D ($N \geq 1$)
- Run system, giving automatic summary X
- What percentage of the n-grams from the reference summaries appear in X?

$$\boxed{ROUGE-2 = \frac{\sum_{S\in\{RefSums\}}\sum_{bi-gram\in S} Count_{match}(bi-gram)}{\sum_{S\in\{RefSums\}}\sum_{bi-gram\in S} Count(bi-gram)}}$$

It is like RECALL:
For all ref summaries - how many bigrams in system generated summary / by
- How many bigrams are present in system summary

If bigram appears in system generated - ROUGE will be higher

ROUGE Example
Reference Summaries
Human 1: water spinach is a green leafy vegetable grown in the tropics.
bigram = no. of words -1 = 10 bigrams
Human 2: water spinach is a semi-aquatic tropical plant grown as a vegetable.
Human 3: water spinach is a commonly eaten leaf vegetable of Asia

System Summary
water spinach is a leaf vegetable commonly eaten in tropical areas of Asia.

Denominator:
How many bigrams are common with summary (bigram = no. of words -1) = 10+10+9
Numerator:
Common bigrams in human-1 + Common bigrams in human-2 + Common bigrams in human-3

ROUGE-2 score $\frac{3+3+6}{10+10+9} = 12/29 = 0.413$

Computer for all document in your corpus

Toolkit available for download.
- It can compute ROUGE score and confidence interval
- Can know how much scores are different

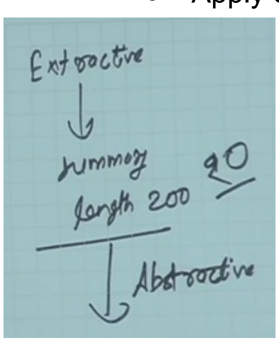
I want summary of length
- Stop at 105 or 107 instead of just length upto 100 (1st 100 words can be taken)
- I want to use stemming
    - Boy and boys will start matching
    - I will remove stop word (is and a)

Define manually some semantic content
- **should have max coverage**

Can you find some property if it is a good summary

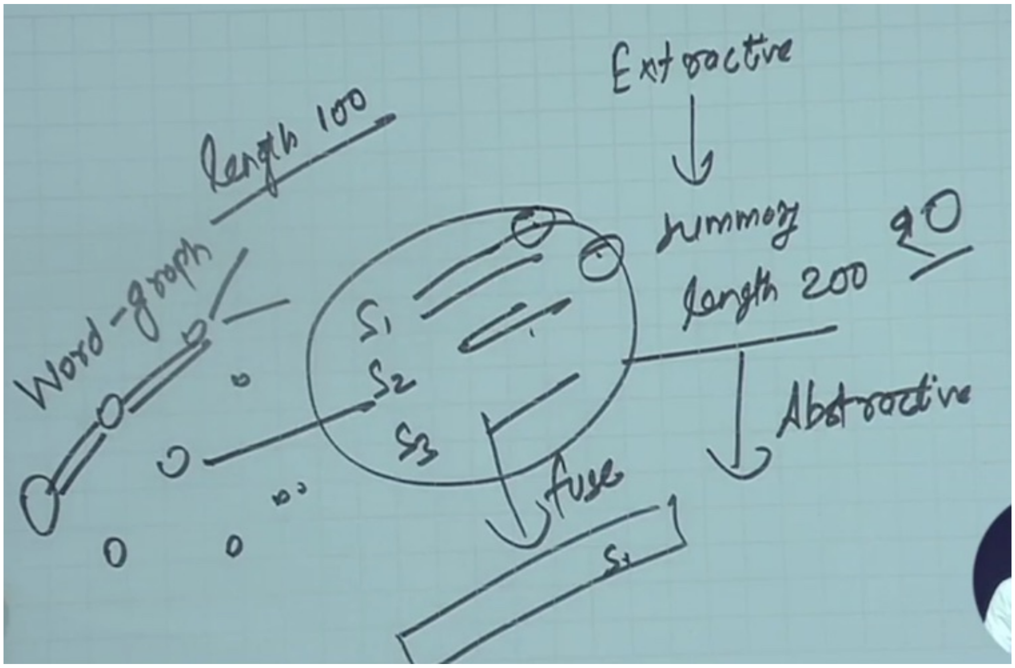
We focused on Generic and single document summarisation

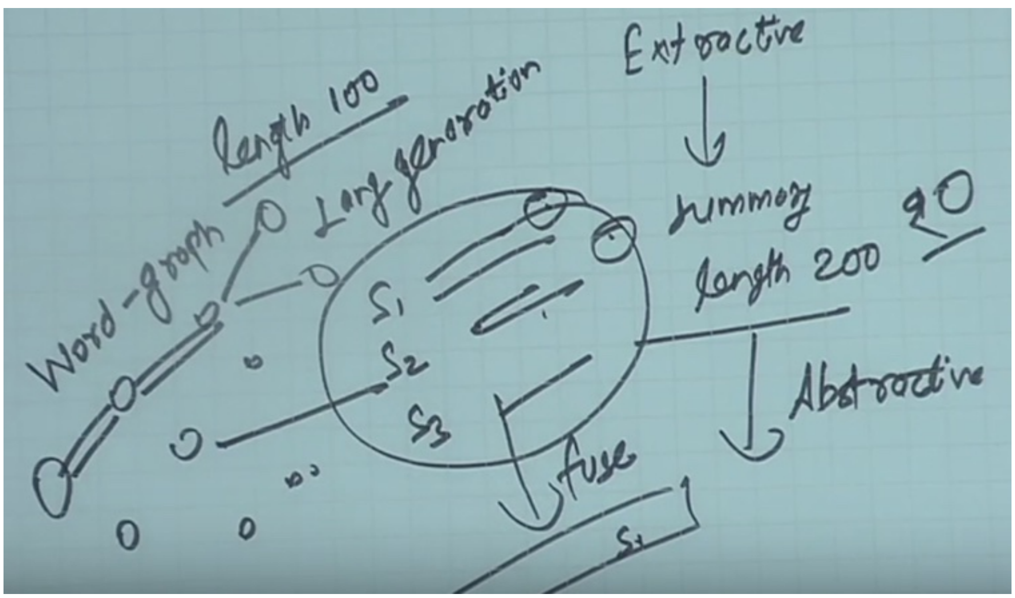
Further Discussions
- Multi-document summarization
    - Can apply same algorithm - LexRank
    - You want to give different weightage to different documents
    - DIfferent way of similarity for
        - Statements with in documents and across documents
    - Overall idea is same
- Query-specific summarization
    - Find out sentences similar to query, then do summarisation
    - Do larger summary than you wanted
    - Then find out which ones are more closer to the query
    - Can do query specific multi document query
        - First select documents, then sentences which are similar
- Abstractive summarization
    - Slightly trickier
    - You are fusing information
    - Apply extractive summary, get a larger set (say summary of length 200)



s1, s2 and s3 are similar, fuse them
Words are common or some information provided by each sentence

Construct a graph





Wherever diverse use generative methods

- Paths can be based on unigram, bigram

Applications:

- Scientific articles
- Twitter stream
- Quora, stackover flow answers
- Disaster events

w11L54: W11L4: Text Classification - I

Example: Positive or negative movie review?

- unbelievably disappointing
- Full of zany characters and richly applied satire, and some great plot twists
- this is the greatest screwball comedy ever filmed
- It was pathetic. The worst part about it was the boxing scenes.

Example:

Male or Female Author? What is the age of author (20s, 30s, old age)
Which country, who is the author?
Authors will have some trait

1. By 1925 present-day Vietnam was divided into three parts under French colonial rule. The southern region embracing Saigon and the Mekong delta was the colony of Cochin-China; the central area with its imperial capital at Hue was the protectorate of Annam…
2. Clara never failed to be astonished by the extraordinary felicity of her own name. She found it hard to trust herself to the mercy of fate, which had managed over the years to convert her greatest shame into one of her greatest assets…

Male - write about different fact
Female - opinion and subjective information

Example: What is the subject of this article? Or Categories of Article

MEDLINE Article

MeSH Subject Category Hierarchy

- Antogonists and Inhibitors
- Blood Supply
- Chemistry
- Drug Therapy
- Embryology
- Epidemiology
- …

Text Classification

- Assigning subject categories, topics, or genres - category of text classification
- Spam detection
  - Email, Tweet, Movie review, Product review - is it spam or not
- Authorship identification
- Age/gender identification
- Language identification
  - English vs Hindi - Hindi in Devanagari - use roman script
  - They have same script - need to go to word level
  - Same translation - pure - english. pure (poore - whole in hindi)
- Sentiment analysis
  -

Text classification: problem definition

d = piece of text, Each document belongs to one class

Input

- A document d
- A fixed set of classes $C = \{c_1, c_2, \ldots, c_n\}$

Output

A predicted class $c \in C$

1. Simplest method:

Classification Methods: Hand-coded rules

- Rules based on combinations of words or other features

Spam

black-list-address OR ("dollars" AND "have been selected")

Pros and Cons

Accuracy can be high if rules carefully refined by expert, *but building and maintaining these rules is expensive.*

Need to have sufficient rule, but will not know how many

Will not have high recall

2. Classification Methods: Supervised Machine Learning

- Naive Bayes
- Logistic regression (LR) - similar to MaxEnt - what is the probability of a class given input x
    - Linear function over various features (exponent of lambda - si)
- Support-vector machines (SVM)
- Decision Tree (DT, RF, XGBoost) ...

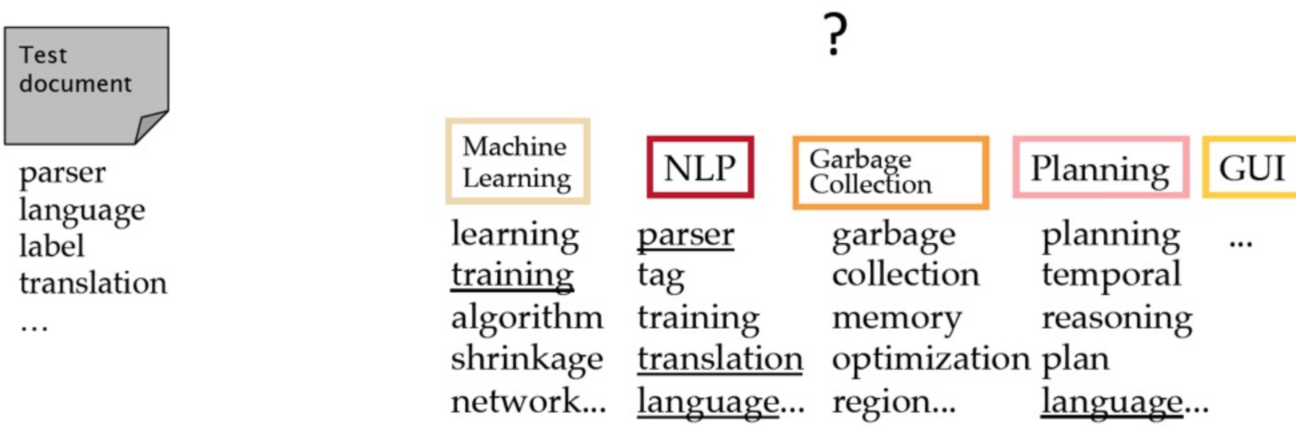
Naive Bayes - Powerful baseline

Naive Bayes Intuition

- Simple classification method based on Bayes' rule
- Relies on very simple representation of document - Bag of words

Bag of words for document classification

- Set of words. Order of the words don't matter



Document = bag or words   Document is assigned class

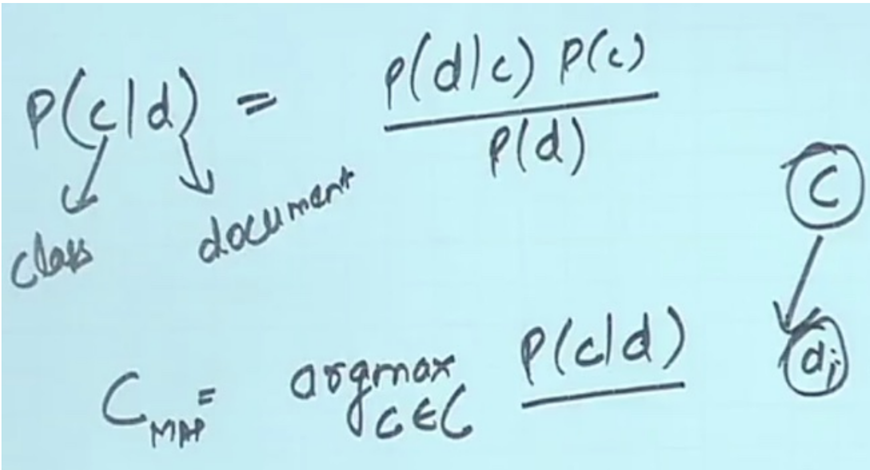
At test time (left) you need to find class of document

Bayes' rule for documents and classes

For a document ***d*** and a class ***c*** $P(c\|d) = \frac{P(d\|c)P(c)}{P(d)}$ Find prob of all different classes Find which one has max prob	***Naive Bayes Classifier*** $c_{MAP} = \underset{c \in C}{\operatorname{arg\,max}} P(c\|d)$ $= \underset{c \in C}{\operatorname{arg\,max}} P(d\|c)P(c)$ $= \underset{c \in C}{\operatorname{arg\,max}} P(x_1, x_2, \ldots, x_n\|c)P(c)$

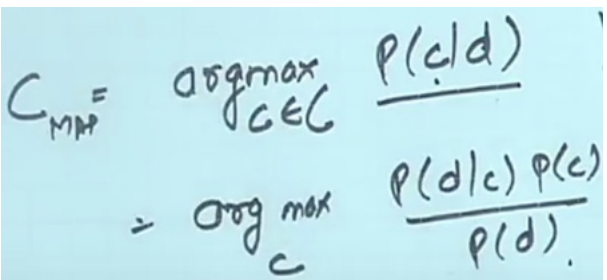
How do you compute P(c|d)

- Generative - have class then generate the document
- Do you want to generate +ve or -ve review



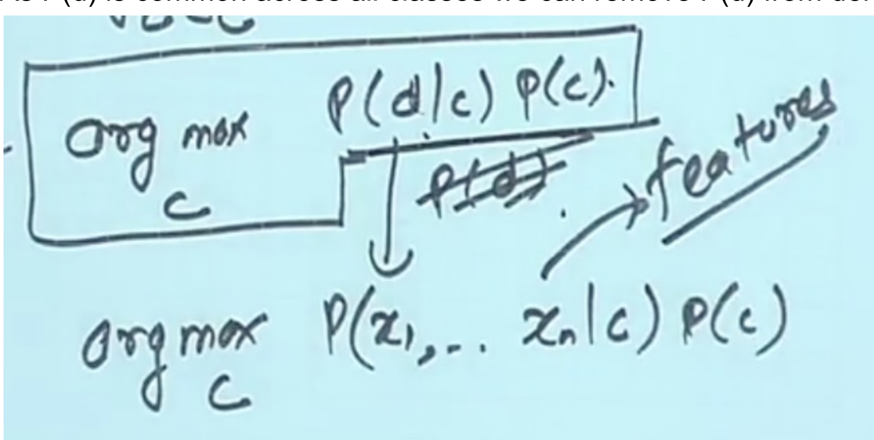
Hence can't compute P(c|d) directly

Use Bayes rule



P(d) is common for all classes

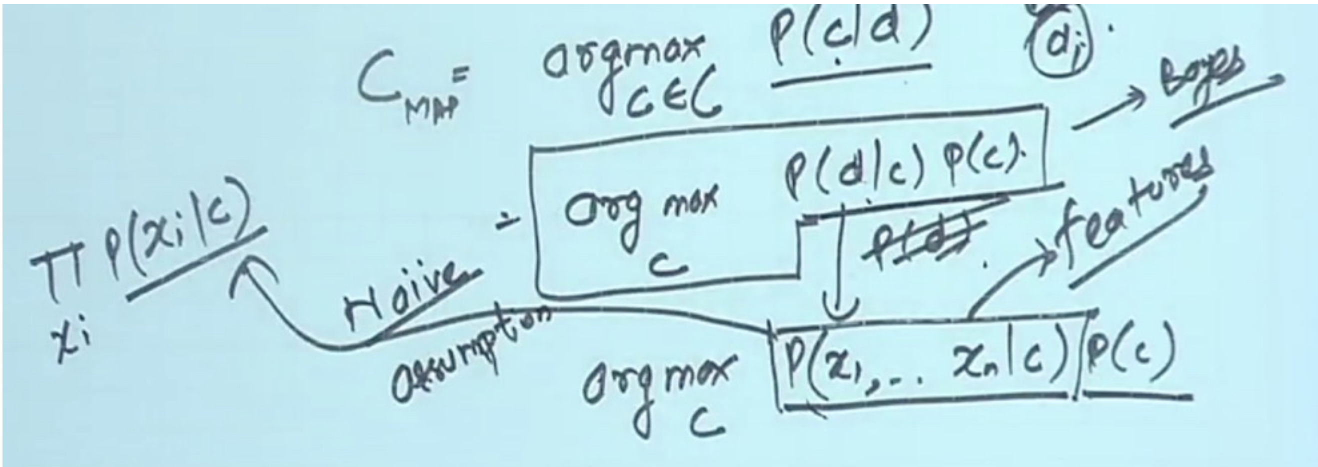
As P(d) is common across all classes we can remove P(d) from denominator



P(c) - prior prob of class - how often class c occurs
P(d|c) - given data what is the probability

- What is your document - bag of words x1,
- X can be words, time of document etc.

What is the probability of getting all features



Don't worry about join probability !!!

$P(x_1, x_2, \ldots, x_n|c)$ **Naive Bayes classification assumptions**

1. Bag of words assumption

Assume that the **position** of a word in the document doesn't matter

2. Conditional Independence

Assume the feature probabilities $P(x_i|c_j)$ are independent given the class c_j.

$$P(x_1, x_2, \ldots, x_n|c) = P(x_1|c) \cdot P(x_2|c) \ldots P(x_n|c)$$

$$c_{NB} = \arg\max_{c \in C} P(c) \prod_{x \in X} P(x|c)$$

X - different words

How do you compute probabilities from training data?

Learning the model parameters

Maximum Likelihood Estimate - P(c) what is the probability of this class in data

$$\hat{P}(c_j) = \frac{doc-count(C = c_j)}{N_{doc}}$$

How many documents have been labelled with this class / no. of documents in training data

$$\hat{P}(w_i|c_j) = \frac{count(w_i, c_j)}{\sum_{w \in V} count(w, c_j)}$$

Prob of word given class = no. of time this words occurs in cj / count of all words in the class cj

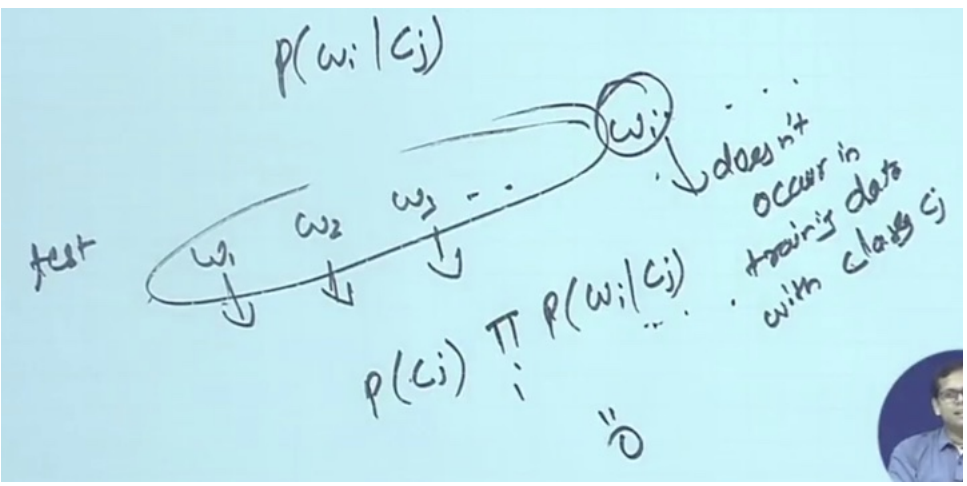
- *Different tokens*

Problem with MLE

Suppose in the training data, we haven't seen the word "*fantastic*", classified in the topic 'positive'.

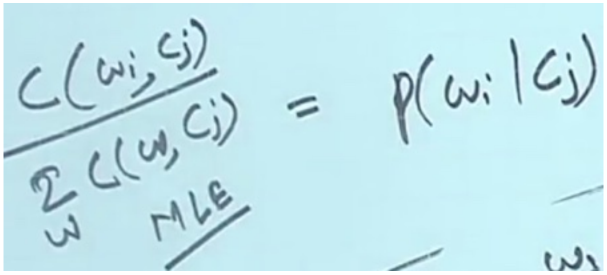
$$\hat{P}(fantastic|positive) = 0$$

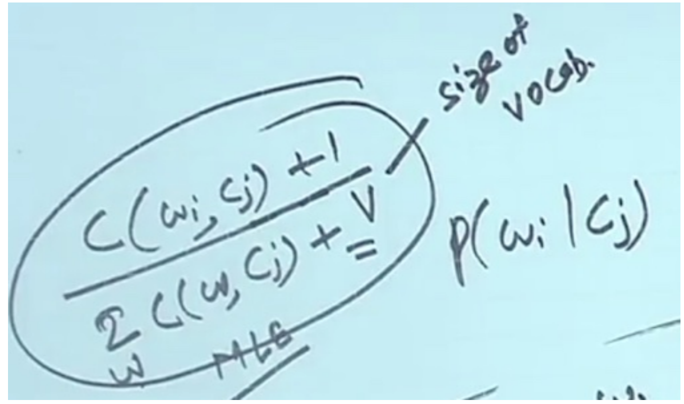
P(wi | cj)



Only one word will make the prob = 0

$$c_{NB} = \arg\max_{c} \hat{P}(c) \prod_{x \in X} \hat{P}(x_i|c)$$





+1 in numerator
+V in denominator

Laplace (add-1) smoothing

$$\hat{P}(w_i|c) = \frac{count(w_i, c) + 1}{\sum_{w \in V} (count(w, c) + 1)}$$

$$= \frac{count(w_i, c) + 1}{(\sum_{w \in V} (count(w, c)) + |V|}$$

So whole thing will not be zero

w11L55: W11L5: Text Classification - II

A worked example

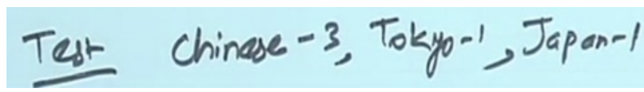
$$\hat{P}(c) = \frac{N_c}{N}$$

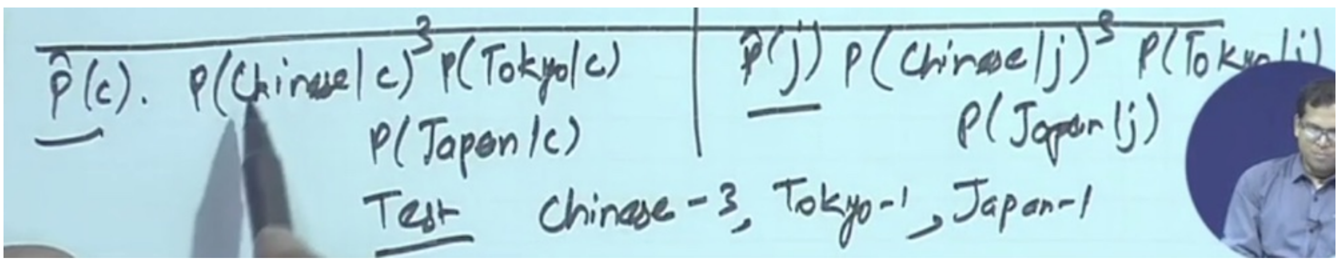
$$\hat{P}(w \mid c) = \frac{count(w,c)+1}{count(c)+|V|}$$

	Doc	Words	Class
Training	1	Chinese Beijing Chinese	c
	2	Chinese Chinese Shanghai	c
	3	Chinese Macao	c
	4	Tokyo Japan Chinese	j
Test	5	Chinese Chinese Chinese Tokyo Japan	?

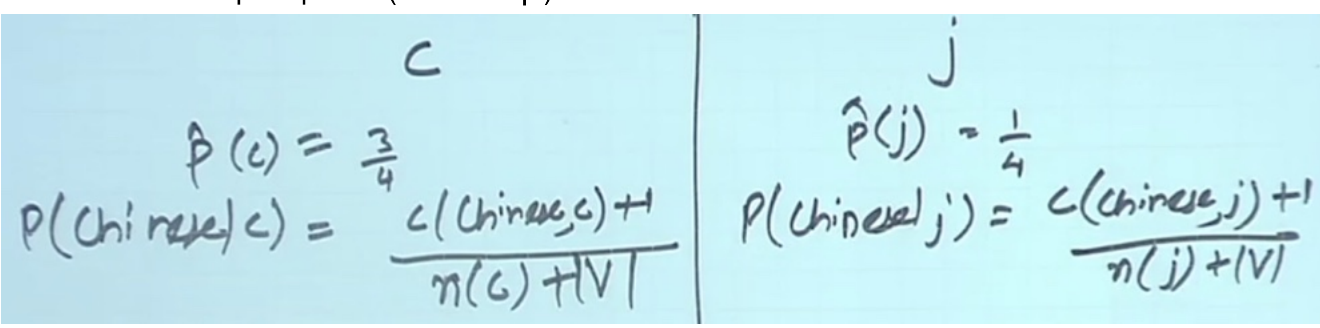
Class c occurs 3 times out of 4

Want to assign test to some class:

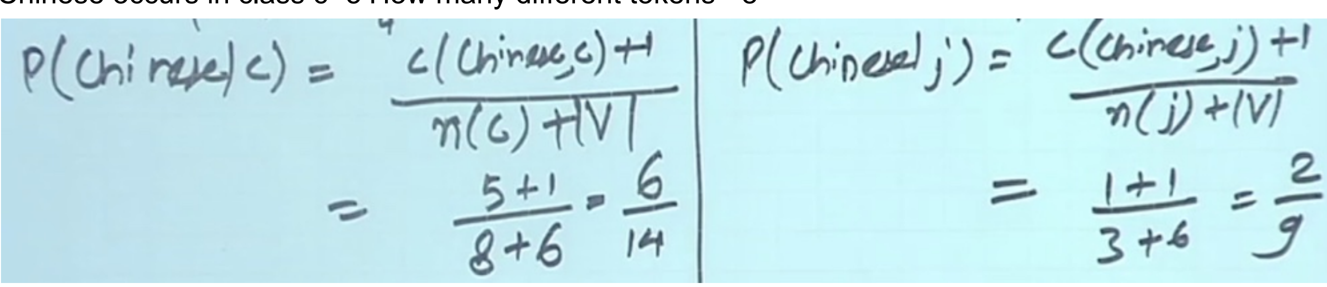




How to compute prob P(chineses |c) =

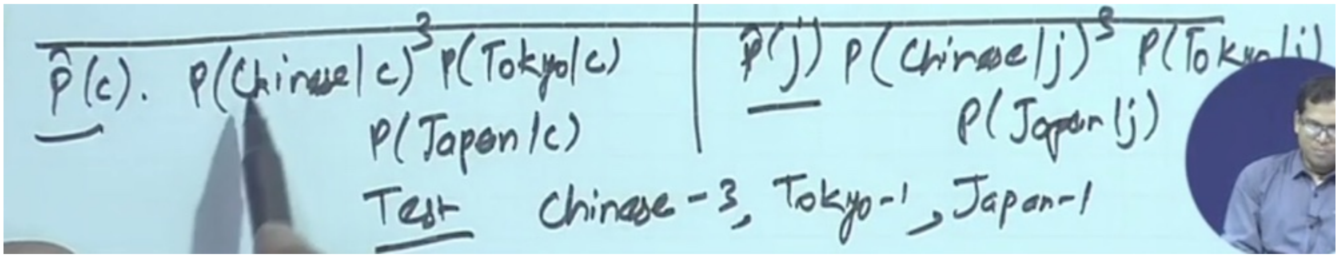


Chinese occurs in class c -5 How many different tokens - 8



P(chinese | c) 3/7   P(chinese | j) 2/9
Computer P(tokyo | c) 1/14 and P (japan | c) 1/14 P(tokyo | j) 2/9 and P (japan | j) 2/9

Compute P(c | d5)   Compute P(j | d5)



$P(c|d5) \propto 3/4 * (3/7)^3 * 1/14 * 1/14 \approx 0.0003$

$P(j|d5) \propto 1/4 * (2/9)^3 * 2/9 * 2/9 \approx 0.0001$

Class C has higher prob than class j

Priors:

$P(c) = \frac{3}{4}$

$P(j) = \frac{1}{4}$

Conditional Probabilities:

P(Chinese|*c*) = (5+1) / (8+6) = 6/14 = 3/7
P(Tokyo|*c*) = (0+1) / (8+6) = 1/14
P(Japan|*c*) = (0+1) / (8+6) = 1/14
P(Chinese|*j*) = (1+1) / (3+6) = 2/9
P(Tokyo|*j*) = (1+1) / (3+6) = 2/9
P(Japan|*j*) = (1+1) / (3+6) = 2/9

Choosing a class:

$P(c|d5) \propto 3/4 * (3/7)^3 * 1/14 * 1/14 \approx 0.0003$

$P(j|d5) \propto 1/4 * (2/9)^3 * 2/9 * 2/9 \approx 0.0001$

Naive Bayes and Language Modeling

In general, NB classifier can use any feature

- URL, email addresses, dictionaries, network features (e.g. email from this person)

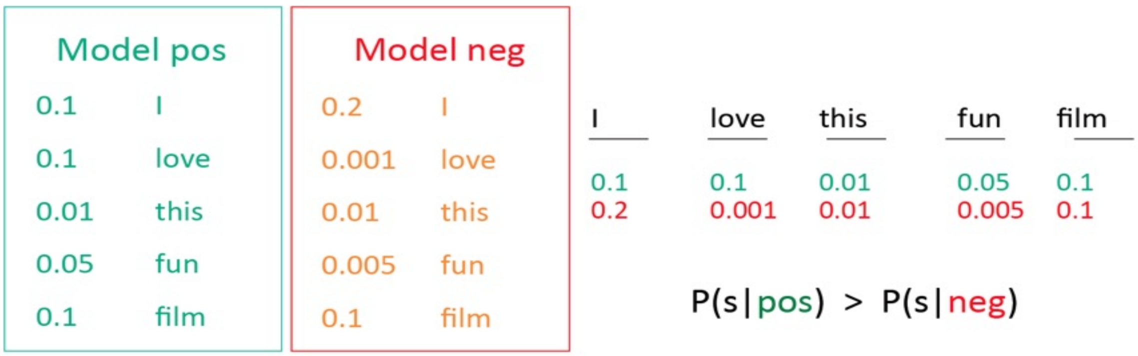
But if we use only the ***word*** *features and all the words in the text*

- Naive Bayes has an important **similarity** to language modeling.
- *Each* ***class*** *can be thought of as a separate* ***unigram*** *language model.*
- *For each class you are building Language Model*
- *Which language model assigns higher probability*

Naive Bayes as Language Modeling

- For +ve class - build a unigram model
- For -ve class - build a unigram model

Which class assigns a higher probability to the sentence?

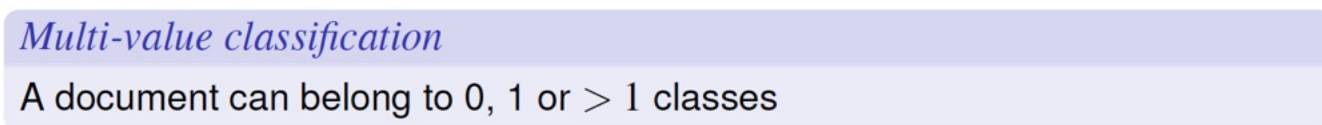


At test time you see word : i love this fun film. +ve model is giving higher probability
Another way of thinking about this problem

Naive Bayes: More than Two Classes

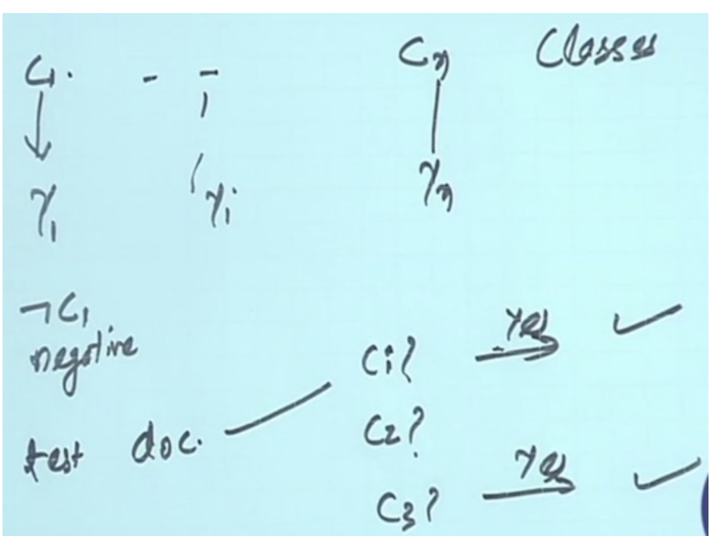
Multi-value classification

- Categories are not mutually exclusive



Make different binary classifier for each class

C1 is positive example,  is negative



Each document at test time, might be given multiple labels

Handling Multi-value classification

- For each class $c \in C$, build a classifier γ_c to distinguish c from all other classes $c' \in C$
- Given test-doc d, evaluate it for membership in each class using each γ_c
- d belongs to any class for which γ_c returns true

Multi-class classification using Binary Classification

One-of or multinomial classification

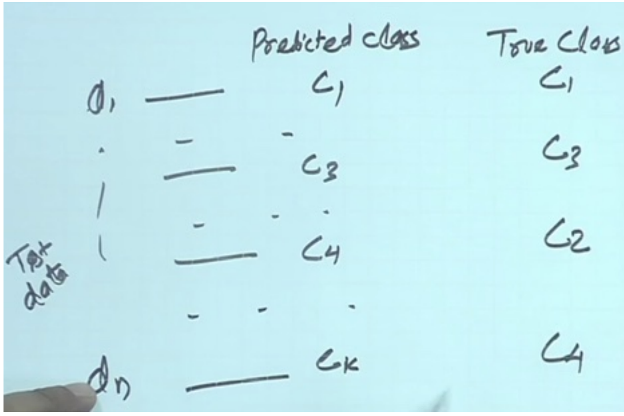
Classes are mutually exclusive: each document in exactly one class

Binary classifiers may also be used

- For each class $c \in C$, build a classifier γ_c to distinguish c from all other classes $c' \in C$
- Given test-doc d, evaluate it for membership in each class using each γ_c
- d belongs to one class with maximum score

What prob the classifier gave -> take the max probability -> assign that class

Evaluation:



Evaluation measures

Accuracy: What fraction of time same answer

Evaluation: Constructing Confusion matrix c

For each pair of classes $< c_1, c_2 >$ how many documents from c_1 were incorrectly assigned to c_2? (when $c_2 \neq c_1$)

Docs in test set	Assigned UK	Assigned poultry	Assigned wheat	Assigned coffee	Assigned interest	Assigned trade
True UK	95	1	13	0	1	0
True poultry	0	1	0	0	0	0
True wheat	10	90	0	1	0	0
True coffee	0	0	0	34	3	7
True interest	-	1	2	13	26	5
True trade	0	0	2	14	5	10

95 = assigned 95 they were true - classifier confused between UK and wheat
90 = poultry to wheat - really confused

How many assigned UK 95+10 (col)
True UK = 110 (row)

Precision: of all assigned UK, 95 were correct

Recall: of UK class - recalled 95 out of 110

Precision Micro	Precision Macro

Per class evaluation measures

Recall

Fraction of docs in class i classified correctly: $\frac{c_{ii}}{\sum_j c_{ij}}$

Cii / Add row-ij

Precision

Fraction of docs assigned class i that are actually about class i: $\frac{c_{ii}}{\sum_i c_{ji}}$

cii/ add col-ji

Accuracy

Fraction of docs classified correctly: $\frac{\sum_i c_{ii}}{N}$

Add diagonal element / total number of docs

This is for individual classes

Micro- vs. Macro-Average
If we have more than one class, how do we **combine multiple performance measures** into **one quantity**?

Macro-averaging
Compute performance (precision, accuracy) for each class, then **average ((0.7 +0.9) /2)**

Micro-averaging
Collect decisions for all the classes, compute contingency table, evaluate.

Micro- vs. Macro-Average

Class 1

	Truth: yes	Truth: no
Classifier: yes	10	10
Classifier: no	10	970

Class 2

	Truth: yes	Truth: no
Classifier: yes	90	10
Classifier: no	10	890

Micro Ave. Table

	Truth: yes	Truth: no
Classifier: yes	100	20
Classifier: no	20	1860

Precision for class1: 10/20 = 0.5 Precision for class2 90/100 = 0.9
Macro-averaged precision: (0.5 + 0.9)/2 = 0.7
Given equal weight

Compute on 3rd table:
Micro-averaged precision: 100/120 = 0.83
Biased towards having more no. of instances

Micro-averaged score is dominated by score on common classes

Use Macro - for imbalance dataset

Resample:

- Undersample
- Oversample (from minority class)
- Or do both
-